

DOTTECH SOFTWARE

QA & SOFTWARE TESTING

INTERNSHIP ASSESSMENT

TEST PLAN, TEST CASES & BUG REPORTS

Prepared By

Safiya Ameen

Higher National Diploma in Information Technology (HNDIT)
Sri Lanka Institute of Advanced Technological Education (SLIATE)

Assessment Track

QA & Software Testing

Features Under Test

User Login
Product Search

Document Type

Software Testing Documentation

Submission Date

September 2026

Testing Basis:

Feature specification provided by DotTech Software

Testing Approach:

Requirements-based manual testing

Confidentiality:

Prepared for the DotTech Software Internship Assessment.

TEST PLAN

Project: DotTech Software Internship Assessment

Track: QA & Software Testing

Features: User Login & Product Search

Prepared By: Safiya Ameen

Testing Approach: Requirements-Based Manual Testing

1. Document Purpose

This Test Plan defines the testing approach for the User Login and Product Search features provided in the DotTech Software internship assessment.

The purpose is to define what will be tested, how it will be tested, the required test data and environment, and the conditions for completing the testing activity.

2. Testing Objectives

The main objectives of testing are to:

- Verify that valid users can log in successfully and reach the dashboard.
- Verify that unsuccessful login attempts display the required error message.
- Verify account locking after three consecutive failed attempts and the specified five-minute lock period.
- Verify email-format validation and the minimum eight-character password requirement.
- Verify that Forgot Password navigates to the password-reset page.
- Verify that product search works after the specified 500 ms period of user inactivity.
- Verify case-insensitive searching and trimming of leading and trailing spaces.
- Verify that search results display the product name, price, and category.
- Verify the required no-results message.
- Verify that clearing the search restores the full product list.
- Identify potential defects that could affect the correct behaviour of the two features.

3. Testing Basis

Testing is based on the feature specification provided by DotTech Software for the User Login and Product Search features.

The specification defines the expected login behaviour, validation rules, account-locking behaviour, search behaviour, and required user-facing results.

No executable application or dedicated test environment was provided with the assessment. Therefore, the test cases represent planned test coverage based on the supplied requirements. The three bug reports are hypothetical sample defects that could be identified during actual test execution.

4. Scope

The following functionality is included in the testing scope.

Feature	In-Scope Areas
User Login	Entering email and password; successful login; dashboard redirection; failed-login error; three consecutive failed attempts; five-minute account lock; lock message; email validation; minimum eight-character password; Forgot Password navigation.
Product Search	Search bar; 500 ms debounce behaviour; search results; product name, price and category; no-results message; case-insensitive search; trimming spaces; clearing search and restoring the full product list.

5. Out of Scope

The following areas are outside the scope because they are not defined in the supplied feature specification.

Area	Reason
Database implementation	Database behaviour is not specified.
Backend/API implementation	API details are not provided.
Password-reset functionality	Only navigation to the password-reset page is specified.
Detailed UI design and styling	The specification mainly defines functional behaviour.
Load and stress testing	No load or performance requirements are provided.
Security testing beyond specified login behaviour	Detailed security requirements are not provided.
Cross-browser and cross-device testing	No browser or device matrix is specified.
Accessibility testing	Accessibility requirements are not provided.

6. Test Types and Why They Are Relevant

Test Type	Why It Is Relevant
Functional Testing	Verifies that the features work according to the stated requirements.
Positive Testing	Verifies expected behaviour with valid inputs and actions.
Negative Testing	Verifies how invalid inputs and unsuccessful actions are handled.
Boundary Testing	Verifies limits such as the eight-character password requirement and three failed login attempts.
Edge Case Testing	Verifies less common inputs such as different letter casing and leading or trailing spaces.
Timing Testing	Verifies the required 500 ms search debounce behaviour.
State/Sequence Testing	Verifies behaviour across a sequence of actions, such as repeated failed logins and clearing a search.

7. Test Approach

Testing will follow a requirements-based approach.

Step	Approach
1	Review the supplied feature specification.
2	Identify testable requirements and test conditions.
3	Create test cases covering positive, negative, boundary and edge scenarios.
4	Prioritize test cases based on feature impact.
5	Execute the test cases when an executable application is available.
6	Compare actual results with expected results.
7	Record identified defects using structured bug reports.
8	Review test coverage against the supplied requirements.

As no executable application was supplied, actual execution results cannot be claimed in this assessment.

8. Test Data

Feature	Test Data
User Login	Valid email; invalid email; correct credentials; incorrect password; passwords below, at and above eight characters; repeated failed login attempts.
Product Search	Existing product terms; non-existing product terms; different letter casing; terms with leading/trailing spaces; empty search; search terms entered with different typing speeds.

Actual credentials and product values would be provided by the test environment during execution.

9. Environment Assumptions

The assessment does not provide an executable application, test URL, browser matrix, device matrix, or dedicated test environment.

For actual test execution, the following would be required:

- A working build or test environment containing the specified features.
- Test user credentials or a method to create test accounts.
- Representative product data.
- A supported web browser and test device.
- A method for recording actual results and defects.

Any browser, device, operating system, or environment requirements not provided in the specification should be confirmed before execution.

10. Entry and Exit Considerations

Entry Criteria

Testing execution should begin when:

- The requirements are available and understood.
- A testable application build or test environment is available.
- Required test accounts and product data are available.
- The environment is accessible and stable enough for testing.

Exit Criteria

Testing may be considered complete when:

- All planned in-scope test cases have been executed.
- Required functional, negative, boundary, edge, timing and state coverage has been addressed.
- Identified defects have been documented and reviewed.
- No unresolved critical issues remain without an agreed decision.
- Test results and defects have been documented.

For this assessment, these are execution criteria rather than completed results because no executable application was supplied.

11. Risks and Limitations

Risks

Risk	Potential Impact
Ambiguity in the feature specification	Different interpretations may lead to different expected behaviours.
Incorrect login attempt handling	Users may be locked or allowed to continue incorrectly.
Incorrect debounce implementation	Search may occur too early, too late, or more often than expected.
Incomplete input handling	Search or validation may behave inconsistently for some inputs.

Limitations

- No executable application or test environment was provided.
- Browser, operating system and device requirements were not specified.
- The exact login error and account-lock message wording is not defined.
- The specification does not state whether a successful login resets the consecutive failed-attempt counter.
- Password rules beyond the eight-character minimum are not specified.
- Database, API, detailed security, accessibility and broader performance requirements are not provided.

12. Deliverables

The following deliverables will be provided for the QA & Software Testing assessment:

Deliverable	Description
Test Plan	Defines the testing scope, objectives, approach, assumptions, risks and limitations.
Test Cases	Minimum 15 detailed test cases covering the User Login and Product Search requirements.
Bug Reports	Exactly 3 sample bug reports based on hypothetical defects that could be identified during execution.
Final QA Document	Consolidated PDF/Word/Markdown document containing the required testing documentation.
Optional Automation	Automation may be included if time permits and will remain an additional bonus activity.

TEST CASES

The following test cases are derived from the User Login and Product Search requirements provided in the DotTech Software internship assessment.

3.1 User Login Test Cases

ID	Title	Preconditions	Test Steps	Expected Result	Priority
TC-LOGIN-001	Login with valid credentials	A registered user account exists and the login page is available.	1. Enter a valid email. 2. Enter the correct password. 3. Submit the login form.	The user logs in successfully and is redirected to the dashboard.	High
TC-LOGIN-002	Login with incorrect password	A registered user account exists.	1. Enter a valid email. 2. Enter an incorrect password. 3. Submit the form.	Login fails and an error message is displayed below the form.	High
TC-LOGIN-003	Login with invalid email format	Login page is available.	1. Enter an invalid email format. 2. Enter a password. 3. Submit the form.	The invalid email is rejected and the user cannot complete a successful login.	High
TC-LOGIN-004	Password with 7 characters	Login page is available.	1. Enter a valid email. 2. Enter a 7-character password. 3. Submit the form.	The password does not meet the minimum eight-character requirement and the login is not accepted.	High
TC-LOGIN-005	Password with exactly 8 characters	Login page is available and an account can use an 8-character password.	1. Enter a valid email. 2. Enter an 8-character password. 3. Submit the form.	The password meets the specified minimum length requirement.	High
TC-LOGIN-006	Password with 9 characters	Login page is available.	1. Enter a valid email. 2. Enter a 9-character password. 3. Submit the form.	The password is accepted as meeting the minimum length requirement.	Medium

ID	Title	Preconditions	Test Steps	Expected Result	Priority
TC-LOGIN-007	Account locks after three consecutive failed attempts	A registered user account exists and is not locked.	1. Enter an incorrect password and submit. 2. Repeat the failed login twice. 3. Observe the result after the third consecutive failure.	After the third consecutive failed attempt, the account is locked and the lock message is displayed.	Critical
TC-LOGIN-008	Login attempt while account is locked	The account has been locked after three consecutive failed attempts.	1. Attempt to log in using the account during the five-minute lock period.	The account remains locked and the user cannot successfully log in during the lock period.	Critical
TC-LOGIN-009	Account becomes available after five-minute lock period	The account has been locked for five minutes.	1. Wait until the five-minute lock period has elapsed. 2. Attempt to log in using valid credentials.	The account is no longer blocked by the specified lock period and a valid login can proceed.	High
TC-LOGIN-010	Forgot Password navigation	Login page is available.	1. Select the Forgot Password option.	The user is navigated to the password-reset page.	Medium

3.2 Product Search Test Cases

ID	Title	Preconditions	Test Steps	Expected Result	Priority
TC-SEARCH-001	Search for an existing product	Product listing page is available and contains a known product.	1. Enter the name of an existing product in the search bar. 2. Stop typing.	Matching products are displayed after the specified search behaviour is triggered.	High
TC-SEARCH-002	Verify product search result information	Search results contain matching products.	1. Search for an existing product. 2. Review the displayed result.	Each search result displays the product name, price, and category.	High
TC-SEARCH-003	Search for a product with no matches	Product listing page is available.	1. Enter a term that does not match any product. 2. Stop typing.	The message “No results found for [term]” is displayed.	High

ID	Title	Preconditions	Test Steps	Expected Result	Priority
TC-SEARCH-004	Search is case-insensitive	A product exists with a known name.	1. Search using the product name in lowercase. 2. Repeat using uppercase or different letter casing.	The search returns the relevant matching product regardless of letter casing.	High
TC-SEARCH-005	Search with leading spaces	A matching product exists.	1. Enter the search term with spaces before the term. 2. Stop typing.	Leading spaces are trimmed and the relevant search results are returned.	Medium
TC-SEARCH-006	Search with trailing spaces	A matching product exists.	1. Enter the search term with spaces after the term. 2. Stop typing.	Trailing spaces are trimmed and the relevant search results are returned.	Medium
TC-SEARCH-007	Search with leading and trailing spaces	A matching product exists.	1. Enter the search term with spaces before and after it. 2. Stop typing.	Leading and trailing spaces are trimmed and the relevant search results are returned.	Medium
TC-SEARCH-008	Verify 500 ms search debounce	Product listing page is available.	1. Enter a search term. 2. Observe the search behaviour before 500 ms. 3. Stop typing and observe after 500 ms.	The search is triggered after the user has stopped typing for the specified 500 ms period.	High
TC-SEARCH-009	Search while typing continuously	Product listing page is available.	1. Begin entering a search term. 2. Continue typing without stopping for 500 ms. 3. Observe the search behaviour.	Search is not triggered prematurely while the user is still typing; it occurs after the required period of inactivity.	High
TC-SEARCH-010	Clear search and restore full product list	Product listing page contains multiple products.	1. Enter a valid search term. 2. Allow the search results to appear. 3. Clear the search field.	The full product list is restored after the search is cleared.	High

3.3 Test Case Coverage Summary

Feature	Number of Test Cases	Coverage
User Login	10	Functional, positive, negative, boundary, state/sequence
Product Search	10	Functional, negative, edge, timing, state
Total	20	Requirements-based coverage

BUG REPORTS

Important Note

The application and test environment were not provided with the feature specification. Therefore, the following are **hypothetical sample bug reports** created from possible defects that could be identified when executing the test cases against an actual application.

These reports are not presented as bugs actually observed in a live application.

BUG REPORT 01

Field	Details
Bug ID	BUG-LOGIN-001
Title	Account is not locked after 3 consecutive failed login attempts
Feature	User Login
Environment	Test environment not provided. Browser, operating system and application build would be recorded during actual testing.
Severity	High
Precondition	A valid user account exists and the login page is available.
Steps to Reproduce	1. Open the login page. 2. Enter a valid email address and an incorrect password. 3. Submit the login form. 4. Repeat the failed login attempt two more times using the same account. 5. Attempt to log in again.
Expected Result	After 3 consecutive failed login attempts, the account should be locked for 5 minutes and the specified lock message should be displayed.
Actual Result	Hypothetical defect: The account remains available for login after the third consecutive failed attempt, instead of being locked for 5 minutes.
Severity Reason	The specified account-locking behaviour is not working, which may affect the protection of user accounts from repeated failed login attempts.
Attachments	No attachment available because the application was not provided. During actual testing, a screenshot or screen recording showing the third failed attempt and the missing lock behaviour would be attached.

BUG REPORT 02

Field	Details
Bug ID	BUG-SEARCH-001
Title	Product search is case-sensitive
Feature	Product Search
Environment	Test environment not provided. Browser, operating system and application build would be recorded during actual testing.
Severity	Medium
Precondition	The product listing page contains a product that can be found using a search term.
Steps to Reproduce	1. Open the product listing page. 2. Enter a product search term using lowercase letters. 3. Note the displayed results. 4. Clear the search field. 5. Enter the same search term using uppercase or different letter casing. 6. Compare the results.
Expected Result	The search should be case-insensitive. The same relevant products should be returned regardless of whether the search term is entered using uppercase or lowercase letters.
Actual Result	Hypothetical defect: The search returns different results when the same search term is entered with different letter casing.
Severity Reason	Users may receive different search results for the same search term because of letter casing, which does not match the specified search behaviour.
Attachments	No attachment available because the application was not provided. During actual testing, screenshots showing the different results for the same term with different casing would be attached.

BUG REPORT 03

Field	Details
Bug ID	BUG-SEARCH-002
Title	Clearing the search field does not restore the full product list
Feature	Product Search
Environment	Test environment not provided. Browser, operating system and application build would be recorded during actual testing.
Severity	High
Precondition	The product listing page contains multiple products.
Steps to Reproduce	1. Open the product listing page. 2. Enter a search term that returns one or more matching products. 3. Wait for the search results to be displayed. 4. Clear the search field completely. 5. Observe the product listing.
Expected Result	After the search field is cleared, the full product list should be restored.
Actual Result	Hypothetical defect: After clearing the search field, the previously filtered results remain displayed instead of restoring the full product list.
Severity Reason	Users cannot return to the complete product list after performing a search, which affects the expected search functionality.
Attachments	No attachment available because the application was not provided. During actual testing, screenshots showing the filtered results and the list after clearing the search field would be attached.

BUG REPORT SUMMARY

Bug ID	Feature	Defect	Severity	Status
BUG-LOGIN-001	User Login	Account does not lock after 3 consecutive failed attempts	High	Hypothetical
BUG-SEARCH-001	Product Search	Search is case-sensitive	Medium	Hypothetical
BUG-SEARCH-002	Product Search	Clearing search does not restore the full product list	High	Hypothetical

Note on Actual Testing

These reports demonstrate how potential defects would be documented if the application were available for execution. Since no executable application or test environment was provided, no actual defect discovery is being claimed.